Security Finding Report: Nmap Assessment of 127.0.0.1

Date of assessment: 5 November 2026
Tool: Nmap 7.993
Target: localhost (127.0.0.1), Windows 11 workstation
Also probed: 192.168.1.0/24 (host discovery only)

1. Executive Summary

A series of Nmap scans was run against the local machine to identify exposed services, fingerprint the operating system, and flag weak configurations. Three TCP ports were found open: **135 (MSRPC)**, **445 (SMB)** and **3306 (MySQL 8.0.40)**. The operating system was identified as **Windows 11 23H2**.

No critical issues were identified. The most significant findings are that **SMB message signing is not required**, the **MySQL version appears outdated**, and the **OS build may be past its servicing window**. Several lower-risk hardening items were also noted. Because all service scans targeted the loopback address, the results show what is listening on the host but do not prove whether these services are reachable from the network.

2. Scope and Methodology

#	Command	Purpose	Result
1	`nmap -sn 192.168.1.0/24`	Host discovery (ping scan)	256 IPs scanned, 0 hosts up (206.88 s)
2	`nmap 127.0.0.1`	Default port scan (top 1000 TCP ports)	3 open ports, 997 closed (0.33 s)
3	`nmap -O 127.0.0.1`	OS detection	Windows 11 23H2, 0 hops (1.40 s)
4	`nmap -sC 127.0.0.1`	Default NSE scripts	MySQL info, TLS certificate, SMB security mode (16.73 s)
5	`nmap -sV 127.0.0.1`	Service/version detection	MSRPC, MySQL 8.0.40, SMB not identified (6.66 s)

#	Command	Purpose	Result
6	`nmap -sV -sC 127.0.0.1 -oN nmap_report.txt`	Combined scan, saved to file	Consistent with scans 4 and 5 (15.96 s)

3. Summary of Findings

ID	Finding	Port	Severity
F-01	SMB message signing enabled but not required	445/tcp	Medium
F-02	MySQL 8.0.40 likely outdated / on an end-of-life release line	3306/tcp	Medium
F-03	Windows 11 23H2 may be out of support	n/a	Medium
F-04	MySQL database service exposed; network reachability unverified	3306/tcp	Low
F-05	MySQL uses an auto-generated self-signed certificate	3306/tcp	Low
F-06	MySQL advertises LOAD DATA LOCAL capability	3306/tcp	Low
F-07	MSRPC endpoint mapper and SMB exposed	135, 445/tcp	Informational
F-08	Service and OS version disclosure	multiple	Informational

4. Detailed Findings

F-01: SMB message signing not required (Medium)

Evidence: The `smb2-security-mode` script reported SMB dialect 3.1.1 with "Message signing enabled but not required".

Risk: When signing is optional, an attacker positioned on the network can attempt NTLM relay attacks against SMB, forwarding captured authentication to gain access to the host or other systems.

Recommendation: Require SMB signing. Set the Group Policy "Microsoft network server: Digitally sign communications (always)" to Enabled, or run `Set-SmbServerConfiguration -RequireSecuritySignature $true`. Apply the matching client-side policy as well.

F-02: MySQL 8.0.40 likely outdated (Medium)

Evidence: Service detection and the `mysql-info` script both report MySQL **8.0.40** on port 3306.

Risk: 8.0.40 dates from late 2024. Oracle has since shipped multiple quarterly security patches, and the 8.0 series is reported to have reached end of life in April 2026, so it no longer receives fixes. Please confirm against Oracle's current lifecycle page. Running an unpatched database exposes it to known vulnerabilities.

Recommendation: Plan an upgrade to a currently supported MySQL release (8.4 LTS or newer) after testing application compatibility, and back up all databases first.

F-03: Windows 11 23H2 may be out of support (Medium)

Evidence: OS detection reported "Microsoft Windows 11 23H2".

Risk: Microsoft ended servicing for Home and Pro editions of 23H2 in November 2025, and Enterprise/Education editions follow in November 2026. Unsupported builds stop receiving security updates. Risk depends on the installed edition.

Recommendation: Check the edition with `winver`, and upgrade to a currently supported Windows 11 release.

F-04: MySQL service listening; exposure unverified (Low)

Evidence: Port 3306 is open on 127.0.0.1. Scanning the loopback address cannot show whether MySQL is bound only to localhost or to all interfaces.

Risk: If MySQL listens on all interfaces and the firewall allows it, other devices could attempt to log in or brute-force credentials.

Recommendation: Check the `bind-address` setting in `my.ini` and set it to `127.0.0.1` if remote access is not needed. Confirm with `netstat -ano | findstr 3306` and restrict inbound access with Windows Defender Firewall. Use least-privilege accounts and strong passwords, and avoid remote root logins.

F-05: Auto-generated self-signed TLS certificate (Low)

Evidence: The certificate subject is `MySQL_Server_8.0.40_Auto_Generated_Server_Certificate`, valid from 2024-10-25 to 2034-10-23.

Risk: A self-signed certificate cannot be verified by clients, so they may skip validation, leaving connections open to man-in-the-middle attacks. The long validity period also means a compromised key stays usable for years.

Recommendation: Where remote connections are used, install a certificate from a trusted or internal CA, require TLS (`require_secure_transport=ON`), and have clients verify the server certificate.

F-06: LOAD DATA LOCAL capability advertised (Low)

Evidence: The capability `SupportsLoadDataLocal` was listed in `mysql-info`.

Risk: If `local_infile` is enabled, a malicious or compromised server can read files from a connecting client, and it has historically featured in SQL-related attack chains.

Recommendation: Set `local_infile=0` unless the feature is required.

F-07: MSRPC and SMB services exposed (Informational)

Evidence: Ports 135 and 445 are open, which is normal for Windows. The `-sV` scan could not positively identify the SMB service (shown as `microsoft-ds?`).

Risk: These services are common targets for lateral movement and have a long history of vulnerabilities if exposed beyond a trusted network.

Recommendation: Block inbound 135 and 445 at the perimeter and on public network profiles. Disable SMBv1 if still enabled, and keep Windows patched.

F-08: Version disclosure (Informational)

Evidence: Exact versions of MySQL and the OS build were revealed by banners and fingerprinting.

Risk: Precise versions help an attacker select matching exploits.

Recommendation: Limit network access to these services so banners are not visible to untrusted hosts.

5. Additional Observations

- **Host discovery:** The ping sweep of 192.168.1.0/24 found no live hosts after about 207 seconds. This suggests the machine is on a different subnet, or that devices block ICMP or are isolated. It should not be read as proof that the network is empty.
- **Scan coverage:** Only the default top 1000 TCP ports were scanned. UDP ports and the full TCP range (1 to 65535) were not tested.
- **Consistency:** Results were identical across all scans, confirming a stable set of three open ports.

6. Prioritised Recommendations

1. Require SMB signing (F-01).
2. Upgrade MySQL and Windows to supported, patched versions (F-02, F-03).
3. Confirm MySQL is bound to localhost and firewall-restricted (F-04).
4. Replace the self-signed certificate and enforce TLS if remote access is needed (F-05).
5. Disable `local_infile` (F-06).
6. Re-scan the machine's LAN IP address with `nmap -sV -p- <LAN-IP>` to see what other devices can actually reach.

7. Limitations

This assessment is based solely on the five scan outputs provided. The scans targeted the loopback interface, so firewall rules and external reachability were not assessed. No authentication, exploitation or vulnerability-database matching was performed. Severity ratings are indicative and should be adjusted to the system's actual role and data sensitivity. Only test systems you own or have written permission to assess.